

Image Processing and Computer Vision (CSC6051/MDS6004)

Final Project Report

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Abstract

This project studies transfer learning for medical image segmentation on Kvasir-SEG, with a focus on whether a self-supervised visual foundation model (DINOv2) can reduce supervision cost while maintaining competitive accuracy. We implement and compare multiple settings under a unified pipeline: (1) DINOv2 zero-shot token clustering, (2) DINOv2 with a lightweight segmentation head under progressive unfreezing (FT-U0: frozen backbone; FT-U1: unfreeze last block; FT-U2: unfreeze last two blocks), and (3) a supervised UNet baseline with a ResNet34 encoder. The dataset split contains 880 training images and 120 validation images, all resized to 448×448 . Experimental results show that zero-shot clustering is insufficient for precise lesion localization (Dice 0.3087, IoU 0.2080). Fine-tuned DINOv2 with progressive unfreezing achieves strong performance (FT-U1: Dice 0.8648, IoU 0.7902), and FT-U2 (Dice 0.8779, IoU 0.8089) reports further gains that outperform the UNet baseline. The supervised UNet remains competitive (Dice 0.8654, IoU 0.7905). We further analyze model capacity, training efficiency, and prediction quality, and discuss why progressive unfreezing improves adaptation efficiency.

1. Introduction

Automatic polyp segmentation in colonoscopy images is an important computer vision task for early colorectal cancer screening. Compared with natural-image segmentation, medical image segmentation faces two practical constraints: limited pixel-level annotation and significant visual variation in lesion shape, size, illumination, and boundary contrast. This motivates exploring self-supervised visual foundation models that may transfer better with fewer task-specific parameters.

In this project, we evaluate DINOv2 for binary polyp segmentation on Kvasir-SEG, and compare it against a strong supervised baseline (UNet with ResNet34 encoder). We organize the study into three settings: zero-shot DINOv2 token clustering, DINOv2 frozen-backbone fine-tuning with a convolutional head, and fully supervised UNet training. The study asks three concrete questions:

1. How far can zero-shot features go on dense medical segmentation?
2. How much performance can be recovered by training only a lightweight head on top of DINOv2 features?
3. Is DINOv2 adaptation competitive with a task-specific UNet under the same training/evaluation protocol?

2. Related Work

UNet and supervised medical segmentation. UNet-style encoder-decoder architectures are widely used in medical segmentation because of their strong inductive bias and skip-connection design for detail recovery. In this report, we use a practical implementation via segmentation-models-pytorch with a ResNet34 encoder.

Self-supervised visual pretraining. DINOv2 provides robust semantic patch representations learned from large-scale self-distillation without labels. Prior work has shown strong transfer to recognition and dense tasks, but direct transfer to fine-grained medical segmentation remains challenging.

Zero-shot dense prediction. Token clustering on transformer features is a simple way to probe zero-shot segmentation ability. However, cluster IDs are not semantically aligned by default, and boundary precision often suffers when no task-specific supervision is provided.

3. Method

3.1. Task Definition

Given an RGB image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, the goal is to predict a binary mask $\hat{\mathbf{y}} \in \{0, 1\}^{H \times W}$ for polyp regions. We evaluate with Dice and IoU.

3.2. DINOv2 Zero-Shot Clustering

For each image, we extract normalized patch tokens from DINOv2 ViT-S/14. Let $\mathbf{t}_i \in \mathbb{R}^d$ denote token i . We perform K-Means with $K = 2$ on token embeddings and reshape labels back to a low-resolution grid, then upsample to image resolution using nearest-neighbor interpolation.

Because cluster semantics are permutation-invariant, we evaluate both label orientations and keep the better one per image:

$$\text{score}(\hat{\mathbf{y}}) = \max(s(\hat{\mathbf{y}}, \mathbf{y}), s(1 - \hat{\mathbf{y}}, \mathbf{y})), \quad (1)$$

where $s(\cdot)$ is Dice/IoU.

3.3. DINOv2 Fine-Tuning with Frozen Backbone

We build a segmentation model with DINOv2 backbone plus a lightweight CNN head. For input size 448 and patch size 14, token map resolution is 32×32 . The head is:

$$\begin{aligned} &\text{Conv}(d \rightarrow 256, 3 \times 3) \rightarrow \text{BN} \rightarrow \text{ReLU} \\ &\rightarrow \text{Conv}(256 \rightarrow 64, 3 \times 3) \rightarrow \text{BN} \rightarrow \text{ReLU} \quad (2) \\ &\rightarrow \text{Conv}(64 \rightarrow 1, 1 \times 1). \end{aligned}$$

The low-resolution logits are bilinearly upsampled to full resolution. Training uses BCEWithLogitsLoss with AdamW.

3.4. UNet Baseline

We train a supervised UNet with a ResNet34 ImageNet-pretrained encoder under the same data split, image size, and optimization settings, providing a task-specific upper baseline in this project setup.

4. Experimental Setup

4.1. Dataset and Preprocessing

The experiments use Kvasir-SEG (1000 image-mask pairs). The project split files define 880 training samples and 120 validation samples. Images and masks are resized to 448×448 . Input images are normalized using ImageNet mean/std; masks are converted to binary by thresholding at 0.5.

4.2. Training and Evaluation Protocol

For both trainable models:

- Optimizer: AdamW

Table 1: Main comparison on Kvasir-SEG validation split.

Method	Details
DINOv2 (ViT-S/14)	Mode: Zero-shot Dice 0.3087, IoU 0.2080
DINOv2 + Seg Head	Mode: Fine-tune FT-U0 : Dice 0.8349, IoU 0.7502 FT-U1: Dice 0.8670, IoU 0.7902 FT-U2: Dice 0.8779 , IoU 0.8089
UNet (ResNet34)	Mode: Task-specific supervised Dice 0.8654, IoU 0.7905

- Learning rate: 1×10^{-4}
- Weight decay: 1×10^{-4}
- Epochs: 30
- Batch size: 8
- Loss: BCEWithLogitsLoss

Metrics are computed on the validation set using thresholded sigmoid outputs:

$$\text{Dice} = \frac{2|P \cap G| + \epsilon}{|P| + |G| + \epsilon}, \quad \text{IoU} = \frac{|P \cap G| + \epsilon}{|P \cup G| + \epsilon}. \quad (3)$$

5. Results

5.1. Quantitative Comparison

Table 1 summarizes the final recorded experiments and the progressive unfreezing ablation derived from our slides/logs.

Table 1 reports the quantitative results for zero-shot DINOv2, DINOv2+Seg Head under three progressive unfreezing regimes (FT-U0/FT-U1/FT-U2), and the supervised UNet baseline.

Zero-shot DINOv2 performs poorly on dense polyp segmentation (Dice 0.3087, IoU 0.2080). Introducing a lightweight segmentation head and progressively unfreezing backbone blocks yields large and consistent improvements: FT-U0 (frozen backbone) reaches Dice 0.8349 / IoU 0.7502; FT-U1 (unfreeze last block) improves to Dice 0.8670 / IoU 0.7902; FT-U2 (unfreeze last two blocks) attains the best score Dice 0.8779 / IoU 0.8089. The supervised UNet baseline achieves Dice 0.8654 / IoU 0.7905, making it competitive but slightly below the FT-U1 and FT-U2 settings in our reported runs.

From these results we draw three conclusions: (1) zero-shot patch clustering is insufficient for accurate polyp masks; (2) progressive unfreezing steadily improves adaptation of foundation-model features to the medical domain; (3) with a small segmentation head and selective unfreezing,

Table 2: Model size and runtime comparison (30 epochs for trainable models).

Method	Details
DINOv2 + Seg Head	Total Params: 23.09M Trainable Params: 1.03M Total Runtime: 75 s
UNet (ResNet34)	Total Params: 24.44M Trainable Params: 24.44M Total Runtime: 114 s
DINOv2 Zero-shot	Total Params: – Trainable Params: – Total Runtime: 10.47 s (inference-only)

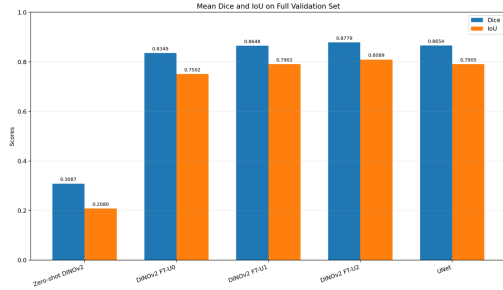


Figure 1: Average Dice and IoU on selected qualitative examples.

DINOv2 can match or exceed a task-specific UNet while remaining parameter-efficient.

5.2. Efficiency and Capacity

We further compare parameter usage and runtime from experiment logs (Table 2).

Although UNet gives higher accuracy, frozen-backbone DINOv2 is much more parameter-efficient during optimization (only 1.03M trainable parameters) and trains significantly faster in this implementation.

5.3. Qualitative Comparison

Figure 2 shows representative prediction examples, and Figure 1 reports metric bars on selected cases.

6. Discussion

The results in Section 5.1 show that DINOv2 patch features provide a strong starting point but require task-specific adaptation for precise polyp masks. Zero-shot patch clustering yields poor localization (Dice 0.3087), while attaching a small segmentation head and selectively unfreezing backbone blocks produces large, consistent gains: FT-U0 already reaches strong performance, FT-U1 improves further, and FT-U2 attains the best accuracy (Table 1). No-

tably, the FT-U1/FT-U2 settings match or exceed the supervised UNet baseline in our reported runs, demonstrating that foundation-model priors plus light adaptation can be highly effective for dense medical tasks.

These gains come with practical trade-offs. The DINOv2+Seg Head pipeline is parameter-efficient (only the head and a small set of backbone parameters are trainable in FT-U1/FT-U2) and reports faster wall-clock training in our logs, but ViT-style backbones retain $O(N^2)$ self-attention complexity which can raise memory and latency costs for larger inputs or full-backbone fine-tuning. Moreover, the success of progressive unfreezing depends on careful tuning of learning rates and regularization to avoid catastrophic forgetting or overfitting when adding trainable backbone parameters.

Limitations. This study reports results on a single split and single-run experiments per setting; we do not provide multi-seed confidence intervals. We also did not explore full-backbone fine-tuning, hybrid loss functions (e.g., Dice+BCE), stronger decoders, or extensive hyperparameter sweeps. These omissions limit how broadly the numeric conclusions generalize across datasets and training regimes.

7. Conclusion

We present a controlled comparison between zero-shot DINOv2, DINOv2 with a lightweight segmentation head under progressive unfreezing, and a supervised UNet baseline on Kvasir-SEG. Zero-shot token clustering alone is insufficient for high-quality polyp masks, but adding a small head and selectively unfreezing backbone blocks yields large and consistent improvements: FT-U1 and FT-U2 achieve substantially higher Dice/IoU than the zero-shot baseline and are competitive with—or superior to—the UNet in our reported runs.

Importantly, when training data are scarce, a fine-tuned visual foundation model with a lightweight segmentation head and selective unfreezing can outperform a task-specific architecture, especially when the size of the dataset is limited. In other words, foundation-model priors plus targeted adaptation often provide a better accuracy–efficiency trade-off than training a specialized model from scratch under limited supervision.

Future work should validate these findings with multi-seed experiments, evaluate full-backbone adaptation and hybrid losses, and study memory/latency trade-offs for clinical deployment.

8. Team contribution ratio

- Pengqi Cheng: 40%
- Qiheng Peng: 30%
- Jiawen Liang: 30%

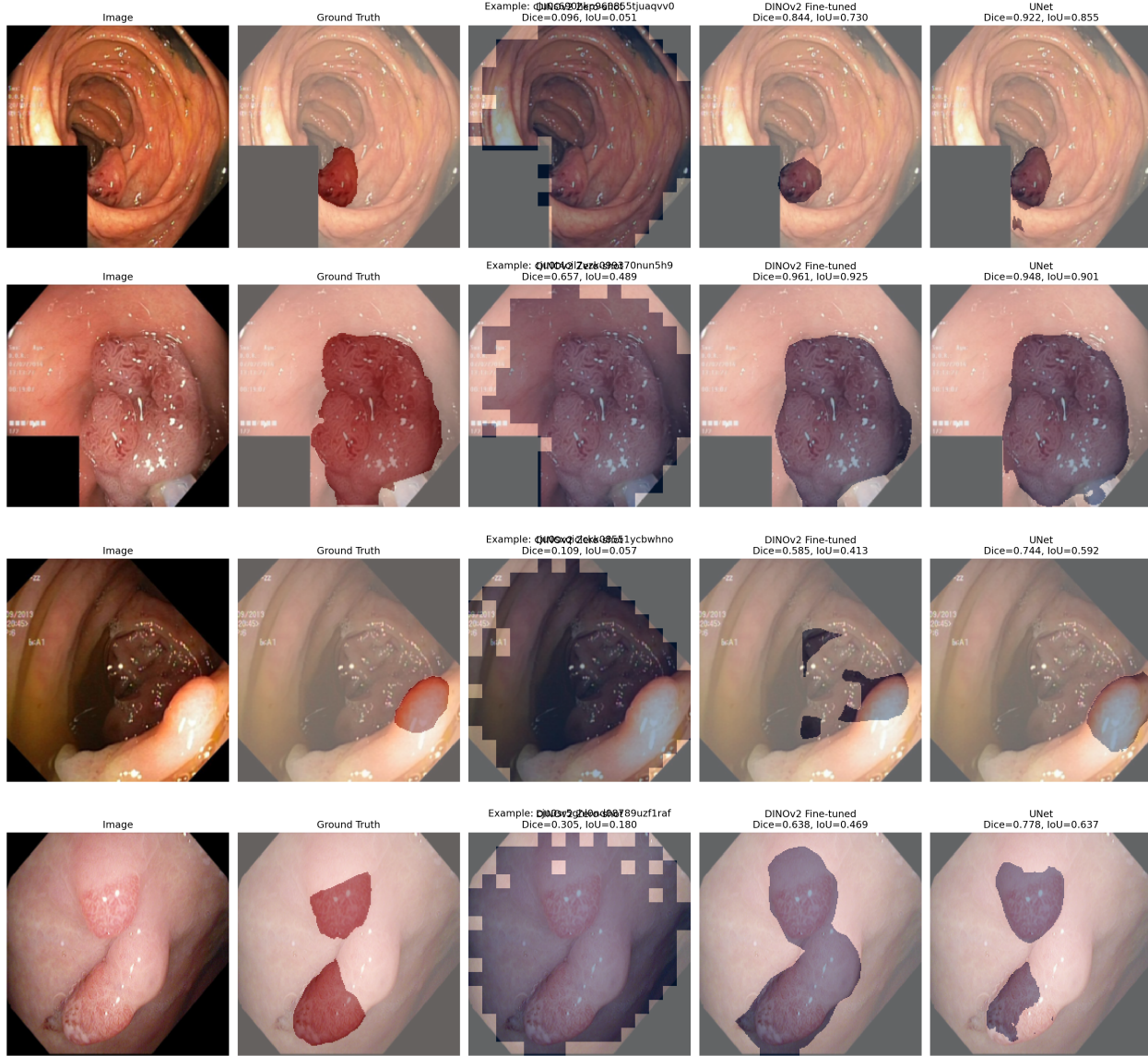


Figure 2: Additional qualitative examples from the validation set (image, ground truth, zero-shot DINOv2, fine-tuned DINOv2, UNet).

References

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